Arithmetic Sequences and Series

Arithmetic sequences

1 Consider the sequence $7, 11, 15, 19, \ldots$

a State the common difference.

b Write a formula for the nth term.

c Find a_{20}.

2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.

a Find the common difference and the first term.

b Write a formula for a_n.

Arithmetic series

3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \cdots$

4 Consider the multiples of 6 strictly between 100 and 500.

a How many are there?

b Find their sum.

5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?

Arithmetic means and sequence properties

6 Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that $5, b_1, b_2, b_3, 25$ is arithmetic).

7 Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.

8 Prove that the sum of the first n positive odd integers $(1 + 3 + 5 + \cdots)$ equals n^2, using the arithmetic series formula.

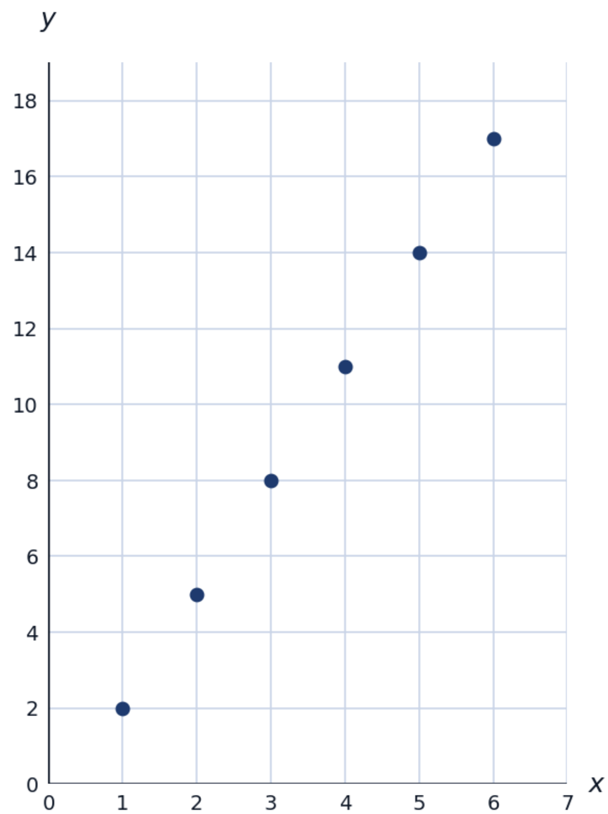
Applications

9 A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

10 A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.

Visualizing an arithmetic sequence

11 The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n, and state the common difference from the graph.



Arithmetic sequences in geometry

12 The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is $80°$ and the angles sum to $540°$ (the pentagon angle sum), find the common difference.

Finding a term given the sum

13 The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n.

Arithmetic sequences with negative common difference

14 A sequence begins $50, 44, 38, 32, \ldots$ Find the first negative term.

Proving an arithmetic sequence property

15 Prove that for any three consecutive terms a_{n-1}, a_n, a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.